24

Lecture 24

The 5G Control Plane

صفحهٔ کنترل نسل پنجم

2G — 3G — 4G — **5G** — 6G

COURSE PART
Part V – Planes and outlook (Lectures 23–26)

DIFFICULTY
Advanced

ESTIMATED STUDY TIME
6–8 hours (3 h reading, 3–5 h exercises)

PREREQUISITES
Lectures 4, 12, 18, 20 and 23

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Original lecture notes for the course *Advanced Communication Networks*. The lecture sequence follows the course syllabus (26 lectures, 2G to 6G). Equations use identical notation in the English and Persian editions. Companion edition: Persian (`lecture-24-fa.pdf`).

1 Learning objectives

After studying this lecture you should be able to:

- **LO1** **Draw** the 5G control-plane protocol stack (RRC, NAS, NGAP, SBI) and **explain** which entities terminate each protocol.
- **LO2** **Describe** the RRC states IDLE, INACTIVE and CONNECTED and their relation to the CM and RM states of the core.
- **LO3** **Trace** a service request from CM-IDLE and **estimate** the control-plane latency of state transitions.
- **LO4** **Compute** the signalling load on an AMF, the inactivity-timer trade-off and the optimal registration-area size.
- **LO5** **Explain** how 5G protects its control plane against overload and signalling storms.

2 Prerequisite knowledge

You need location management and paging (Lecture 4), the MME and its overload control (Lecture 12), the service-based architecture (Lecture 18), the registration and PDU-session procedures (Lecture 20) and the user plane of Lecture 23. This lecture looks at the same system from the signalling side: who talks to whom, in which state, and at what cost.

Key terms. The vocabulary of this lecture with its Persian equivalents:

Term	Persian	Meaning in this lecture
Control plane	صفحهٔ کنترل	Signalling that sets up, maintains and releases connectivity.
RRC	کنترل منابع رادیویی	Radio Resource Control between UE and gNB.
NAS (MM / SM)	لایهٔ غیردسترسی	Non-access stratum between UE and AMF (mobility) or SMF (sessions).
NGAP	پروتکل کاربرد NG	Application protocol on N2 between gNB and AMF, over SCTP.
RRC_INACTIVE	حالت غیرفعال RRC	State in which the UE context is kept but the radio connection is suspended.
CM / RM state	حالت مدیریت اتصال / ثبت	Core view of the UE: connected or idle, registered or deregistered.
Service request	درخواست خدمت	NAS procedure that moves a UE from CM-IDLE to CM-CONNECTED.
Registration area	ناحیهٔ ثبت	List of tracking areas in which an idle UE need not report its location.
RNA	ناحیهٔ اعلان مبتنی بر RAN	Area in which an inactive UE moves without notifying the network.
Signalling storm	طوفان سیگنالینگ	Burst of synchronised procedures that overloads control-plane nodes.

3 Introduction and motivation

The control plane does not carry a single user bit, yet it decides whether user bits can flow at all. Every time a smartphone wakes up to fetch a notification, it runs a small protocol exchange across the radio, the gNB, the AMF, the SMF and the UPF. Multiplied by millions of devices and dozens of transitions per hour, this signalling dominates the processing load of the core and a noticeable share of the radio resources – and its latency is felt by the user as the delay before the first packet.

5G reorganises the control plane in three ways: a new RRC state, RRC_INACTIVE, that avoids most core signalling for short bursts; a clean separation of mobility (AMF) and session (SMF) management; and service-based interfaces inside the core [3], [7]. This lecture explains the resulting protocol stack and state machines and gives the models to dimension them.

4 Fundamental concepts

4.1 The control-plane stack

Three protocol families make up the control plane (Figure 1):

- **RRC** between UE and gNB [4]: system information, connection setup, security activation on the radio, configuration of radio bearers, measurements and handover commands. It runs over PDCP, RLC and MAC on signalling radio bearers.
- **NAS** between UE and core [2]: NAS-MM (registration, service request, authentication, NAS security) terminates in the AMF; NAS-SM (PDU-session establishment and modification) is relayed by the AMF to the SMF. The gNB forwards NAS messages without reading them – they are integrity-protected and ciphered end to end between UE and AMF.
- **NGAP** between gNB and AMF on N2 [1]: UE context management, PDU-session resource setup, paging and handover signalling, over SCTP. Inside the core, the AMF and SMF use service-based interfaces (HTTP/2 with JSON, Lecture 18).

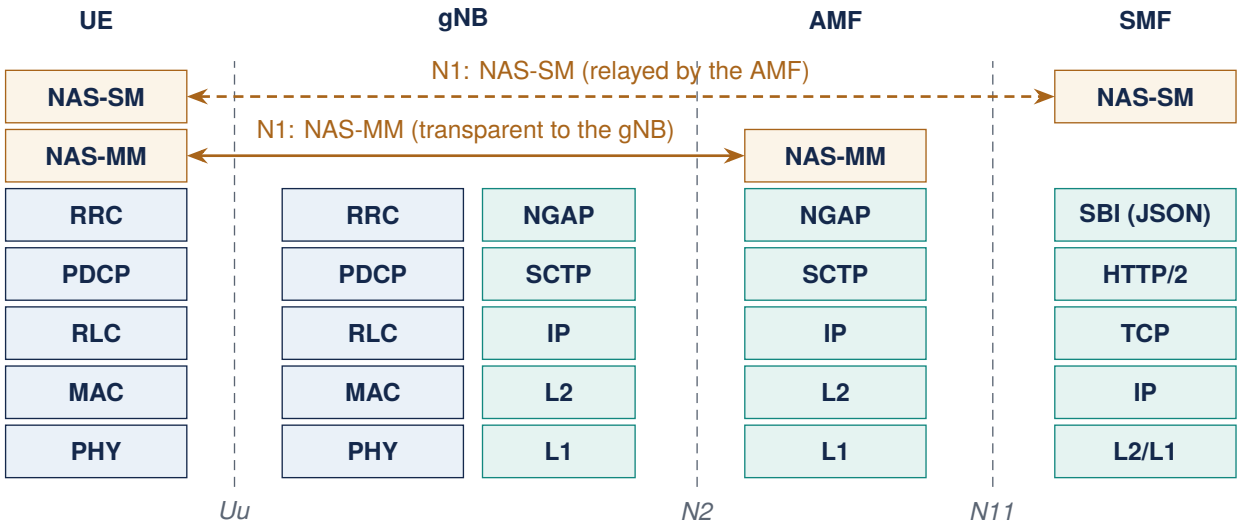


Figure 1. 5G control-plane protocol stack: RRC on Uu, NGAP on N2, NAS end to end between UE and AMF/SMF, and SBI in the core.

4.2 States of a UE

The network keeps three coupled state machines for every UE [4], [7]:

- **RM** (registration management) in the AMF: RM-DEREGISTERED or RM-REGISTERED.
- **CM** (connection management) in the AMF: CM-IDLE (no N2 connection for the UE; location known per registration area) or CM-CONNECTED (an N2 connection exists).
- **RRC** in the gNB and UE: RRC_IDLE, RRC_INACTIVE or RRC_CONNECTED (Figure 2).

In RRC_INACTIVE the UE behaves almost like an idle UE – it camps on cells, monitors paging and moves freely inside a RAN-based notification area (RNA) – but the gNB that suspended it keeps the UE context and the N2/N3 connections to the core remain up, so the core sees the UE as CM-CONNECTED. Returning to RRC_CONNECTED takes an RRC Resume exchange with no core signalling.

4.3 Reachability

Where the network can find the UE depends on its state: in CM-IDLE the AMF knows the registration area (a list of tracking areas) and must page there; in RRC_INACTIVE the anchor gNB knows the RNA and performs RAN paging itself; in RRC_CONNECTED the serving cell is known.

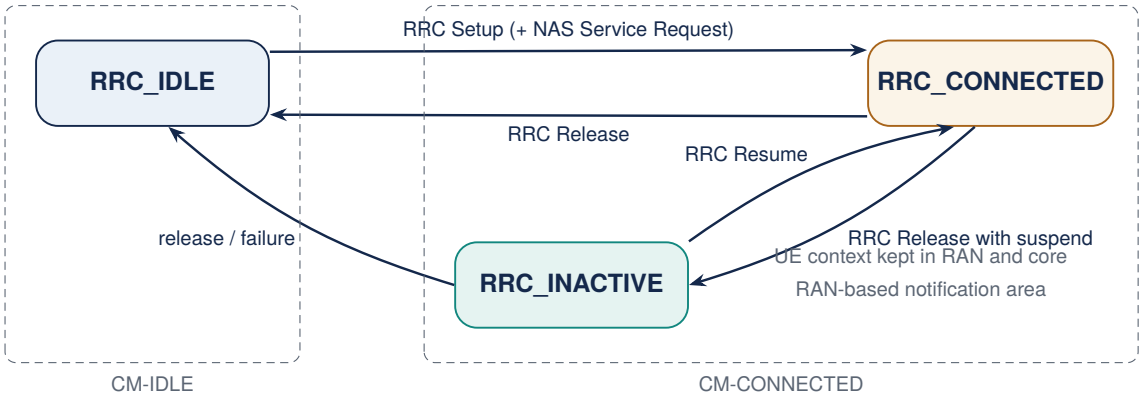


Figure 2. NR RRC states and transitions; RRC_INACTIVE belongs to CM-CONNECTED in the core.

The UE reports leaving its area by a mobility registration update (CM-IDLE) or an RNA update (RRC_INACTIVE), and refreshes it periodically.

5 Detailed technical explanation

5.1 Service request

When an idle UE has uplink data – or is paged for downlink data – it runs the service request (Figure 3) [5]. After random access the UE sets up an RRC connection and piggybacks the NAS Service Request on RRC Setup Complete; the gNB forwards it in an NGAP Initial UE Message to the AMF selected by the 5G-S-TMSI. The AMF checks NAS integrity, asks the SMF of each PDU session to be activated to re-establish its user plane (the SMF updates the UPF over N4), then sends an Initial Context Setup Request carrying the security key and the session resources. The gNB activates AS security, reconfigures the radio bearers and returns its downlink tunnel endpoints, which the SMF installs in the UPF.

5.2 Suspend and resume

With RRC_INACTIVE the gNB ends a connection by an RRC Release with suspend indication: the UE and the gNB keep the AS context and the security configuration, the UE receives a resume identity and an RNA. When the UE has data again it sends an RRC Resume Request; if it is in a cell of another gNB, that gNB retrieves the context from the anchor gNB over Xn and asks the AMF to switch the N2 path. In the common case of resuming in the same area, no NAS or core signalling is needed at all. Small amounts of data can even be sent during the resume procedure itself (small-data transmission, Release 17).

5.3 Separation of mobility and session management

In the EPC the MME handled both mobility and session signalling (Lecture 12). In 5G the AMF handles only mobility, access and NAS security, while the SMF handles sessions. A UE has one AMF but may have several SMFs (one per PDU session or slice). This separation allows non-3GPP access (Wi-Fi via N3IWF) to share the same AMF, keeps the AMF independent of session types, and lets slices have their own SMFs (Lecture 21).

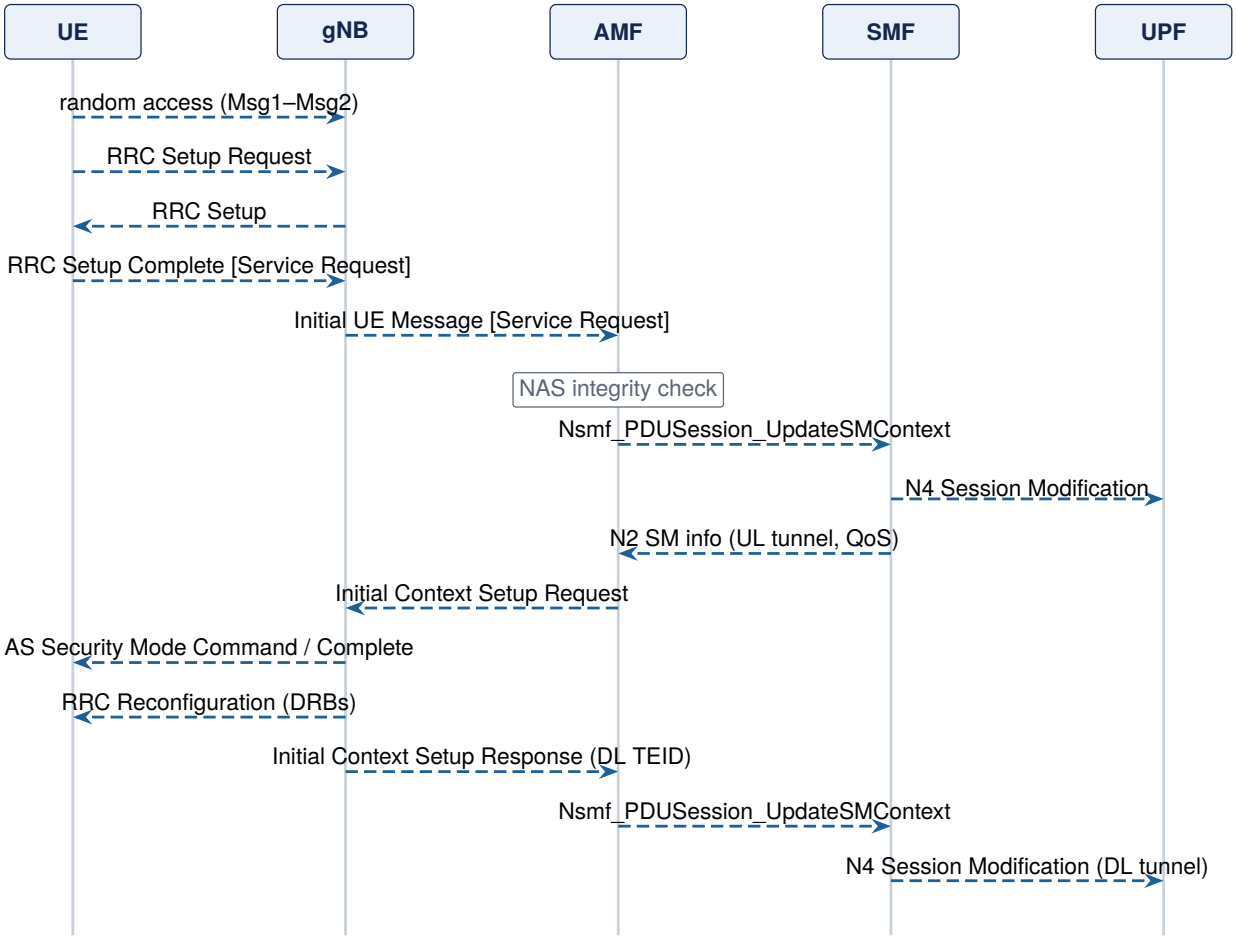


Figure 3. UE-triggered service request from CM-IDLE (simplified, after TS 23.502).

5.4 Overload and congestion control

Control-plane nodes protect themselves at several levels [2], [7]: unified access control in the UE bars access categories broadcast by the gNB before any message is sent; the AMF can ask gNBs to reject or reduce new RRC connections (NGAP overload start); NAS congestion control returns a back-off timer (e.g. T3346) in reject messages so that UEs retry later; and the NRF and SCP spread load over several AMF and SMF instances (Lecture 18).

6 Mathematical models

6.1 Signalling load

Model. If N UEs each perform procedure k at rate λ_k per hour and each procedure costs m_k messages at the AMF, the message rate in messages per second is

$$\Lambda = \frac{N}{3600} \sum_k \lambda_k \, m_k. \tag{1}$$

Example. $N = 2 \times 10^6$ with busy-hour rates per UE: 20 service requests (4 messages each), 2 mobility registrations (6), 3 handovers (6), 0.1 initial registrations (20) and 0.5 PDU-session establishments (8): $\sum_k \lambda_k m_k = 116$ messages per UE per hour and $\Lambda = 64\,444$ messages/s. If RRC_INACTIVE turns 80 % of the idle-to-connected transitions into resumes without core signalling, the sum falls to 52 and $\Lambda = 28\,889$ messages/s – a reduction of 55 %.

6.2 Inactivity timer

Model. A UE generates data bursts as a Poisson process of rate λ. After each burst the gNB keeps the connection for an inactivity time T before releasing (or suspending) it. A transition to idle happens after a burst only if the next burst comes later than T, so the transition rate and the mean connected "tail" per burst are

$$r(T) = \lambda \, e^{-\lambda T}, \qquad \bar{t}_{\text{tail}}(T) = \frac{1 - e^{-\lambda T}}{\lambda}. \tag{2}$$

The fraction of time spent in the tail is $\lambda \, \bar{t}_{\text{tail}} = 1 - e^{-\lambda T}$.

Example. One burst per minute ($\lambda = 1/60\,\text{s}^{-1}$): $T = 10$ s gives 50.8 transitions per hour and 15.4 % of the time in the tail; $T = 30$ s gives 36.4 transitions per hour and 39.3 % (Figure 4). A longer timer saves signalling but costs UE energy and radio resources; RRC_INACTIVE makes short timers affordable because each transition is cheap.

6.3 Registration-area size

Model. With the fluid-flow mobility model (Lecture 4), a UE moving at mean speed v in an area of perimeter P and size A crosses its border at rate $vP/(\pi A)$. For a square area of side a this is $R_u = 4v/(\pi a)$. If each update costs w_u and each paging event (rate λ_p) costs w_p in each of the $n = a^2/A_c$ cells of size A_c, the cost per UE and hour is

$$C(a) = w_u \frac{4v}{\pi a} + w_p \lambda_p \frac{a^2}{A_c}, \tag{3}$$

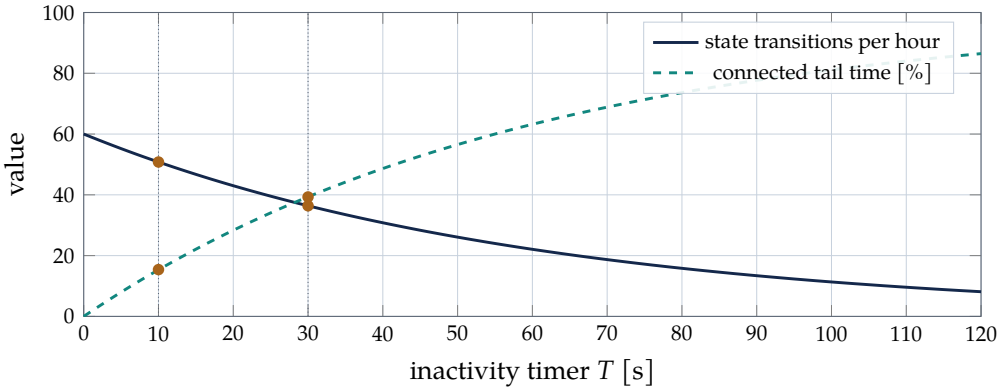


Figure 4. Inactivity-timer trade-off for one burst per minute (Eq. (2)).

minimised by

$$a^* = \left(\frac{2 w_u v A_c}{\pi w_p \lambda_p} \right)^{1/3}. \tag{4}$$

At the optimum the update cost is exactly twice the paging cost.

Example. $v = 20\,\text{km/h}$, $A_c = 1\,\text{km}^2$, $\lambda_p = 5$ per hour, $w_u = 10$, $w_p = 1$: $a^* = 2.94\,\text{km}$ (about 8.7 cells per area), with update cost 86.6 and paging cost 43.3 per UE and hour (Figure 5).

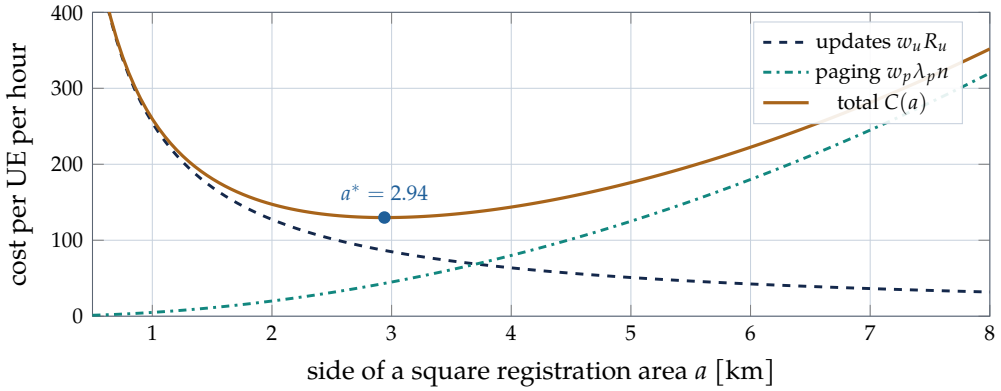


Figure 5. Registration-area size trade-off (Eqs. (3) and (4)).

6.4 Control-plane latency

Model. The time from a battery-efficient state to the start of data transfer is the sum of the steps of the transition:

$$T_{\text{CP}} = T_{\text{RA}} + T_{\text{RRC}} + T_{\text{core}} + T_{\text{ctx}}. \tag{5}$$

TR 38.913 sets a target of 10 ms [6]; ITU-R M.2410 requires 20 ms and encourages 10 ms [9].

Example. With illustrative values for 30 kHz subcarrier spacing – random access 2 ms, RRC exchange 4 ms, core signalling (N2, AMF, SMF and N4) 6 ms, context setup with AS security and reconfiguration 4 ms – the service request from RRC_IDLE takes 16 ms. A resume from RRC_INACTIVE needs only random access and the RRC exchange: 6 ms. Only the inactive state meets the 10 ms target.

7 Network architecture and diagrams

The control plane spans the UE, the gNB (CU-CP after a split, Lecture 23), the AMF set serving a region, and the SMFs, PCF, UDM and AUSF reached through the SBA. AMFs are deployed as sets of interchangeable instances sharing UE contexts in a storage function, so that a gNB can reach any instance of the set (a GUAMI identifies the set and pointer). Figures 1 to 3 show the protocol view, the state view and the procedure view of this architecture.

8 Step-by-step procedures

Procedure 1 • Mobile-terminated data for an idle UE

1. A downlink packet arrives at the UPF for a session without an N3 tunnel; the UPF buffers it (or forwards it to the SMF) and notifies the SMF.
2. The SMF asks the AMF to make the UE reachable (Namf_Communication_N1N2MessageTransfer).
3. The AMF pages the UE in all cells of its registration area through the gNBs (NGAP Paging).
4. The UE answers with a service request (Figure 3); the user plane is re-established and the buffered data are delivered.

Procedure 2 • Resuming from RRC_INACTIVE

1. The UE (with uplink data, or after RAN paging) performs random access and sends an RRC Resume Request with its resume identity.
2. If the context is held by another gNB, the new gNB retrieves it over Xn and later asks the AMF for an N2 path switch.
3. The gNB answers with RRC Resume; security is restored from the stored context with fresh keys.
4. The UE confirms with RRC Resume Complete and data flow on the existing N3 tunnels.

9 Worked examples

Worked example 1 • Dimensioning an AMF set

Problem. For the load of Section 6.1, how many AMF instances are needed if each processes $\mu = 20\,000$ messages/s and should run at no more than 70 % utilisation? Compare with and without RRC_INACTIVE.

Solution. Usable capacity per instance $0.7\mu = 14\,000$ messages/s. Without INACTIVE: $64\,444/14\,000 = 4.6$, so 5 instances. With INACTIVE: $28\,889/14\,000 = 2.06$, so 3 instances. Signalling efficiency in the RAN translates directly into fewer core instances.

Worked example 2 • Choosing the inactivity timer

Problem. With one burst per minute, what is the longest timer that keeps the UE in the tail for at most 20 % of the time, and how many transitions per hour result?
Solution. $1 - e^{-T/60} \leq 0.2 \Rightarrow T \leq -60 \ln 0.8 = 13.4\,\text{s}$. Transitions: $60\, e^{-13.4/60} = 60 \times 0.8 = 48$ per hour.

Worked example 3 • Registration area for fast users

Problem. Repeat the example of Section 6.3 for vehicular users with $v = 60\,\text{km/h}$.
Solution. $a^{*3} = 2 \times 10 \times 60 \times 1/(\pi \times 5) = 76.4$, $a^* = 4.24\,\text{km}$ (about 18 cells). Update cost $10 \times 4 \times 60/(\pi \times 4.24) = 180$, paging cost $5 \times 4.24^2 = 90$, total 270 per UE and hour. Faster users call for larger areas, which is why operators give TA lists along motorways and railways more cells.

10 Technology comparison

Table 1. Control plane in LTE/EPC and in NR/5GC.

	LTE / EPC	NR / 5GC
RRC states	IDLE, CONNECTED (suspend/resume for IoT only)	IDLE, INACTIVE, CONNECTED
Mobility and sessions	MME (both), S-GW/P-GW control	AMF (mobility), SMF (sessions)
Core interfaces	Diameter, GTPv2-C (point to point)	SBI over HTTP/2, NRF discovery
RAN–core protocol	S1AP over SCTP	NGAP over SCTP
Location areas	TA lists, paging by MME	TA lists (CM-IDLE) and RNA with RAN paging (INACTIVE)
Access control	ACB, EAB, back-off timers	Unified access control, back-off timers
Non-3GPP access	Separate ePDG path	Same AMF via N3IWF

Signalling is the price of efficiency. Each idle period saves energy and radio resources but must be paid for by a transition. The 5G control plane lowers that price (INACTIVE, fewer core messages, faster processing), which lets operators release connections earlier and serve many more devices per unit of core capacity.

11 Key takeaways

Key takeaways

- RRC (UE–gNB), NAS-MM (UE–AMF), NAS-SM (UE–SMF via AMF), NGAP (gNB–AMF over SCTP) and SBI in the core.
- RRC_INACTIVE keeps the context in the RAN and the core in CM-CONNECTED: resume needs no core signalling.

- Load $\Lambda = (N/3600)\sum_k \lambda_k m_k$; idle-to-connected transitions dominate it.
- Inactivity timer: transitions $\lambda e^{-\lambda T}$ versus tail fraction $1 - e^{-\lambda T}$.
- Registration area: $a^* = (2w_u v A_c/(\pi w_p \lambda_p))^{1/3}$; updates cost twice the paging at the optimum.

12 Exercises

Exercise 1 • Level 1 – conceptual • Where is the UE?

For each RRC state, give the CM state seen by the AMF and the granularity at which the network knows the UE's location.

Exercise 2 • Level 1 – conceptual • Transparent NAS

Why is NAS transparent to the gNB? Give a security reason and an architectural reason.

Exercise 3 • Level 2 – numerical • AMF load

An AMF set serves 5×10^5 UEs with busy-hour rates per UE of 30 service requests (4 messages each), 1 mobility registration (6) and 4 handovers (6). Compute the message rate.

Exercise 4 • Level 2 – numerical • Chatty applications

A messaging application produces one burst every 20 s on average. Compute the transitions per hour and the tail fraction for $T = 5$ s and $T = 10$ s.

Exercise 5 • Level 3 – analytical • Optimal area

Derive Eq. (4) from Eq. (3), show that the update cost is twice the paging cost at the optimum, and compute a^* for pedestrians ($v = 5$ km/h) with the other values of Section 6.3.

Exercise 6 • Level 3 – analytical • Signalling storm

After a core outage, 10^6 UEs must register again, each registration costing 20 messages at an AMF set of 5 instances with $\mu = 20\,000$ messages/s. If the UEs spread their attempts uniformly over a window W, what is the smallest W for 100 % and for 70 % utilisation? Which mechanisms enforce such a spread?

Solutions

Solution to Exercise 1

RRC_IDLE: CM-IDLE, registration area (list of TAs). RRC_INACTIVE: CM-CONNECTED, RNA known to the anchor gNB (the AMF knows the gNB). RRC_CONNECTED: CM-CONNECTED, serving cell.

Solution to Exercise 2

Security: NAS messages are protected end to end with keys shared only by UE and AMF, so a compromised or untrusted RAN node (including non-3GPP access) cannot read or alter them. Architecture: the RAN is independent of core procedures; new NAS features need no RAN change, and the same NAS runs over 3GPP and non-3GPP access.

Solution to Exercise 3

$\sum_k \lambda_k m_k = 120 + 6 + 24 = 150$ per UE and hour; $\Lambda = 5 \times 10^5 \times 150/3600 = 20\,833$ messages/s.

Solution to Exercise 4

$\lambda = 1/20\,\mathrm{s}^{-1}$, 180 bursts per hour. $T = 5\,\mathrm{s}$: $180\,e^{-0.25} = 140.2$ transitions per hour, tail $1 - e^{-0.25} = 22.1\,\%$. $T = 10\,\mathrm{s}$: $180\,e^{-0.5} = 109.2$, tail 39.3 %. Frequent small bursts are the case for which RRC_INACTIVE and small-data transmission were designed.

Solution to Exercise 5

$C'(a) = -4w_u v/(\pi a^2) + 2w_p \lambda_p a / A_c = 0 \Rightarrow a^3 = 2w_u v A_c/(\pi w_p \lambda_p)$. At the optimum $w_p \lambda_p a^2 / A_c = 2w_u v/(\pi a)$, i.e. paging cost $= 2w_u v/(\pi a)$, while update cost $= 4w_u v/(\pi a)$: twice as much. Pedestrians: $a^{*3} = 2 \times 10 \times 5/(5\pi) = 6.37$, $a^* = 1.85\,\mathrm{km}$.

Solution to Exercise 6

Total 2×10^7 messages; capacity 10^5 messages/s. 100 %: $W \geq 200\,\mathrm{s}$; 70 %: $W \geq 2 \times 10^7/(7 \times 10^4) = 286\,\mathrm{s}$. Mechanisms: randomised back-off timers in NAS rejects (e.g. T3346), unified access control barring broadcast by the gNBs, AMF overload indications to the RAN, and prioritisation of emergency and high-priority access.

13 Further reading

The RRC protocol and states are specified in TS 38.331 [4] and summarised in TS 38.300 [3]; NAS in TS 24.501 [2]; NGAP in TS 38.413 [1]; the CM and RM states in TS 23.501 [7] and the procedures in TS 23.502 [5]. The latency targets come from TR 38.913 [6] and ITU-R M.2410 [9]. Rommer et al. [10] explain the 5G core signalling; Dahlman et al. [8] the RRC states and small-data transmission.

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